

Team Member Contributions

AI-Powered Cafeteria Management System

This document presents the responsibility distribution of the project team. The project contains a FastAPI backend, database models, AI/ML decision-support modules, web dashboard pages, a React Native mobile application, data/training scripts, tests, and documentation. The responsibilities below divide the work into three balanced and technically clear areas.

01 — Cemil Alp

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Cemil Alp

AI/ML Models · Analytics · Decision Support

MAIN RESPONSIBILITIES

- Developed the waste prediction logic and ML-based waste estimation pipeline.
- Worked on menu optimization using popularity, waste risk, nutritional balance, and cost criteria.
- Developed the ML-supported hybrid production planning and portion recommendation logic.
- Contributed to sentiment analysis and feedback analysis from student ratings and comments.
- Implemented anomaly detection, trend analysis, predictive analysis, and alert logic.
- Worked on model tracking, automatic training, experiment evaluation, and XAI explanations.
- Prepared and maintained model training scripts and model metric outputs.
- Evaluated model behavior using metrics, feature importance, and validation results.

RELATED PROJECT FILES

ai_yemekhane/modules/waste_analyzer.py
ai_yemekhane/modules/menu_optimizer.py
ai_yemekhane/modules/production_planner.py
ai_yemekhane/modules/sentiment_analyzer.py
ai_yemekhane/modules/feedback_analyzer.py
ai_yemekhane/modules/anomaly_detector.py
ai_yemekhane/modules/trend_analyzer.py
ai_yemekhane/modules/predictive_analyzer.py
ai_yemekhane/modules/alert_system.py
ai_yemekhane/modules/auto_trainer.py
ai_yemekhane/modules/model_tracker.py
ai_yemekhane/modules/xai_explainer.py
ai_yemekhane/modules/experiment_engine.py
ai_yemekhane/scripts/training/
ai_yemekhane/models_data/

Backend Architecture · Database · API Integration

MAIN RESPONSIBILITIES

- Designed and maintained the FastAPI backend structure.
- Managed the main application entry point and router integration.
- Worked on SQLAlchemy database models and migration structure.
- Implemented and integrated API endpoints for menu, ratings, waste, analytics, production planning, reports, cost analysis, IoT, voting, simulation, notifications, and WebSocket status.
- Connected business logic modules with API routes.
- Supported data import and database preparation workflows.
- Contributed to backend configuration, environment setup, and deployment readiness.

RELATED PROJECT FILES

`ai_yemekhane/main.py`

`ai_yemekhane/models.py`

`ai_yemekhane/config.py`

`ai_yemekhane/api/routes/`

`ai_yemekhane/api/dependencies.py`

`ai_yemekhane/alembic/`

`ai_yemekhane/scripts/data/`

MAIN RESPONSIBILITIES

- Designed and maintained the web dashboard and student-facing interface pages.
- Worked on HTML templates, shared layouts, styling, and JavaScript interactions.
- Developed the React Native mobile application screens for menu display, meal rating, chatbot access, and statistics.
- Integrated the mobile application with backend API endpoints.
- Contributed to PDF/report generation and project documentation.
- Prepared user workflows for menu viewing, rating submission, admin monitoring, production entry, reports, simulations, and voting.
- Supported test coverage for API endpoints, chatbot behavior, sentiment analysis, menu optimization, and waste analysis.
- Contributed to presentation/report materials and final project documentation.

RELATED PROJECT FILES

ai_yemekhane/templates/
ai_yemekhane/static/css/
ai_yemekhane/static/js/
ai_yemekhane/mobile_app/
ai_yemekhane/modules/pdf_report.py
ai_yemekhane/tests/
docs/
README.md

Summary of Division

Team Member	Main Area	Core Contribution
Cemil Alp	AI/ML, analytics, decision support	Developed prediction, optimization, anomaly detection, and explainability logic.
Ali Batuhan Zeytün	Backend, database, API integration	Built and integrated the server-side system that connects data, routes, and business modules.
Muhammet Talha ■at■r	UI, mobile app, reporting, QA	Built user-facing interfaces, mobile screens, reporting outputs, and testing/documentation.

Short Version for the Final Report

The project work was divided into three main responsibility areas. **Cemil Alp** focused on AI/ML development, including waste prediction, menu optimization, production planning, anomaly detection, model tracking, and explainable AI outputs. **Ali Batuhan Zeytün** focused on backend architecture, database modeling, API

integration, and system configuration. **Muhammet Talha ■at■r** focused on the user-facing parts of the project, including web dashboard pages, the React Native mobile application, PDF/reporting outputs, testing, and documentation. This distribution allowed the team to cover the full system lifecycle from intelligent decision support and backend services to user interaction and reporting.